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EGYPT WEATHER & DEMOGRAPHICS

Web Scraping + REST API Data Collection Pipeline | Project 02

1. Project Overview

An end-to-end Python data collection pipeline that combines live weather data from the Open-Meteo API with demographic information scraped from Wikipedia for six major Egyptian cities. The two sources are merged into one clean analytical dataset.

2. Pipeline Architecture

Stage 1 — API Ingestion: Requests are sent to Open-Meteo using latitude and longitude for Cairo, Alexandria, Giza, Luxor, Aswan, and Port Said to collect temperature, wind speed, and observation time.

Stage 2 — Web Scraping: Requests + BeautifulSoup / pandas.read_html are used to inspect the Wikipedia city tables and extract governorate and population information.

Stage 3 — Data Integration: The weather and demographic tables are merged with pandas using the common city field.

Stage 4 — Export: The final DataFrame is saved as egypt_weather_collected.csv and also exported to Excel.

Stage 5 — Verification: The saved CSV is reloaded and inspected using head(), info(), describe(), and isnull().sum().

3. Technology Stack

Python • Requests • Pandas • BeautifulSoup • lxml / pandas.read_html • Jupyter Notebook (VS Code)

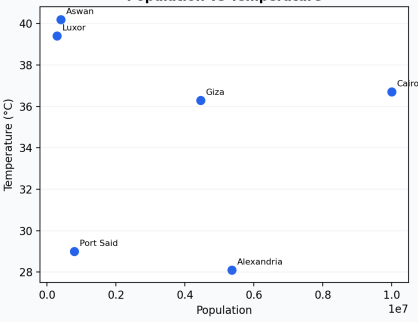
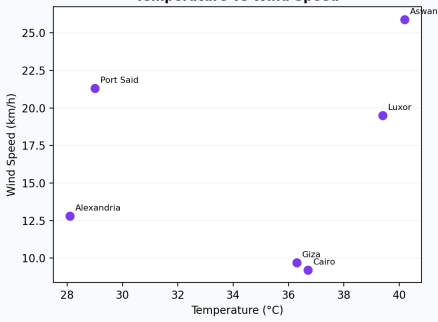
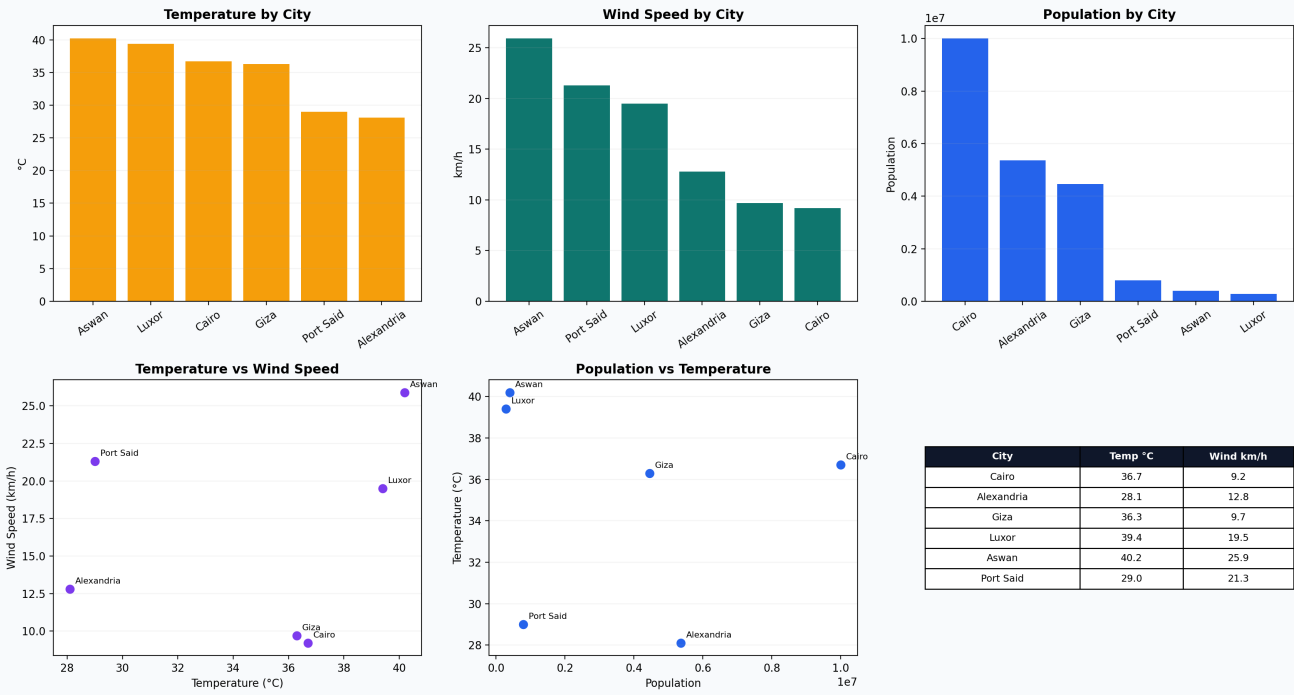
4. Final Dataset

City	Temp °C	Wind km/h	Population
Cairo	36.7	9.2	10,000,000
Alexandria	28.1	12.8	5,362,517
Giza	36.3	9.7	4,458,135
Luxor	39.4	19.5	284,952
Aswan	40.2	25.9	401,890
Port Said	29.0	21.3	791,749

The final pipeline contains 6 rows and 6 attributes: city, temperature_c, windspeed_kmh, observed_at, governorate, and population. The notebook's final validation reports zero missing values across the dataset.

5. Client-Facing Dashboard

EGYPT WEATHER & DEMOGRAPHICS — DATA COLLECTION DASHBOARD



6. Insights from the Collected Data

Temperature: Aswan is the hottest of the six cities at 40.2°C, while Alexandria is the coolest at 28.1°C.

Wind: Aswan also records the highest wind speed at 25.9 km/h, while Cairo has the lowest at 9.2 km/h.

Population: Cairo has the largest population in this collected table at 10,000,000, while Luxor has the smallest at 284,952.

Weather vs Population: The six-city sample does not show an obvious direct relationship between population and temperature. This is an observation from a very small dataset, not a general conclusion about Egypt.

Important limitation: The weather readings represent one observation time in the collected dataset (2026-08-07T15:00). Therefore, the project should be described as a snapshot, not as a long-term climate analysis.

7. What Makes the Project Strong

The project demonstrates a practical data-engineering pattern: collect data from different real-world sources, standardize the structures, join the sources, export the result, reload it, and verify the final dataset.

It also demonstrates that the same Python workflow can combine API data and scraped web data into a single dataset that can later support visualization or further analysis.

8. Skills Demonstrated

REST API Requests • JSON Parsing • Web Scraping • BeautifulSoup • HTML Tables • pandas.read_html • Pandas DataFrames • Data Merging with pd.merge • CSV / Excel Export • Data Validation with head/info/describe/isnull

9. Portfolio Positioning

This case study focuses on Web Scraping and API Data Collection. It does not claim advanced AI techniques because the supplied notebook does not contain a machine-learning or AI model. The strongest evidence in this project is the complete collection-and-integration pipeline.

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